

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

SmartCare Community Clinic currently manages patient information and appointments using spreadsheets, paper records and manual processes. This has caused several problems, including duplicate appointment bookings, difficulty finding patient records, inconsistent appointment statuses and limited visibility of practitioner availability. The clinic also has problems with manual cancellations, unreliable appointment history and producing basic reports. SmartCare needs a simple software system to improve these processes. The first version of the system should focus on managing patients, practitioners and appointments. The system should remain simple and manageable because the clinic does not require a complex hospital information system.

2. Initial stakeholders

Stakeholder	Possible need
Receptionist / clinic staff	Manage patient records, appointments and cancellations
Patients	Have their information and appointments recorded accurately
Healthcare practitioners	View their availability and appointments
Clinic management	Access reliable information and basic operational reports

3. Initial features

Feature	Confirmed or provisional?	Why?
Appointment management	Confirmed	The client specifically wants appointment management
Patient management	Confirmed	Patient management is part of the initial system scope
Practitioner management	Confirmed	Practitioner management is part of the initial system scope
Patient search	Provisional	The clinic has difficulty locating patient records, but the exact search function is not specified

4. Questions for the client

1. What patient information needs to be stored?
2. What practitioner information needs to be stored?
3. What appointment statuses should the system support?
4. How should the system prevent duplicate appointment bookings?
5. What basic operational reports should the system produce?

5. What we do not yet know

1. The exact information that must be stored for patients and practitioners
2. The rules for booking, changing and cancelling appointments
3. The exact types of operational reports management requires

AI Activity Card - Ask, Check, Explain

Before AI

What do I think the code does? What problems can I already identify?

The code stores appointment information, including the patient's name, practitioner name and appointment time. It can add appointments to a list and display the appointments that have been recorded.

AI request

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

1. It does not prevent duplicate appointments
2. It does not properly validate all appointment information
3. Appointments are only stored while the program is running
4. There is no way to cancel or update an appointment
5. Practitioner availability is not checked

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope
Valid blank patient names	yes	no	No	No
Prevent duplicate bookings	yes	No	No	No
Add practitioner availability checking	yes	No	No	No
Add online payments	No	Yes	Yes	Yes

Decide

For each significant suggestion: Accept / Modify / Reject / Keep unverified.

Verify

- Run the code
- Test normal input
- Test unusual input
- Compare with requirements
- Ask tutor/peer

- Check documentation

Explain

Can I explain the final code without reading the AI response? What do I still need to understand?

Yes. I should be able to explain what each part of the code does, including how appointments are stored, how the function adds appointments and how the appointments are displayed. I still need to understand how the program could prevent duplicate bookings, validate different types of incorrect input and manage appointment changes or cancellations.